Menglin Zhu

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Curriculum Vitae with References
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Research Interest

- Probing quantum/energy materials atom by atom using advanced (S)TEM:
  - Multislice electron ptychography ▪ *Operando* microscopy ▪ 4D-STEM ▪ Spectroscopy
- Modeling atomistic behavior and phase transitions using DFT and *ab initio* MD simulations.
- Extracting insights from microscopy data using statistical analysis and machine learning

Education and Training

2023–present **Postdoctoral Researcher**, *Massachusetts Institute of Technology*, Prof. James M. LeBeau.
2018–2023 **Ph.D.in Materials Science and Engineering**, *Ohio State University*, Prof. Jinwoo Hwang.
2014–2017 **B.S. in Materials Science and Engineering**, *Ohio State University*, Prof. Jinwoo Hwang.

Honors & Awards

2026 ***Albert Crewe Award*** of Microscopy Society of America to recognize distinguished contributions to the field of microscopy in the physical sciences of an early career scientist

2025 ***PARADIM user proposal #366*** *Probing Dynamic Polar Structures in Relaxors*, 10 days of microscope time at Cornell PARADIM center

2023 ***Postdoctoral Scholar*** of Microscopy Society of America, awarded as a postdoctoral researcher for the paper submitted for the Microscopy and Microanalysis conference 2023

2022 ***Student Scholar*** of Microscopy Society of America, awarded as a Ph.D. candidate for the paper submitted for the Microscopy and Microanalysis conference 2022

2021 ***Presidential Fellow*** of Ohio State University, the most prestigious award given by the Graduate School to outstanding Ph.D. candidates

2017 ***The Mars Fontana Scholarship*** by Ohio State University, awarded to the most outstanding senior student in the Department of Material Science and Engineering

2016 ***Summer Research Fellow*** of Ohio State University
Markworth-Woolley Scholarship by Ohio State University

Publications

Citations: 1439, h-Index: 17, *Equal Contribution, #Correspondance

[50] *2026* **Menglin Zhu***, Michael Xu*, Zishen Tian*, Colin Gilgenbach, Daniel Drury, Bridget R Denzer, Ching-Che Lin, Deokyoung Kang, Lane W Martin, and James M LeBeau. Operando Electron Microscopy of Nanoscale Electronic Devices on Nonconductive Substrates. *Microscopy and Microanalysis*, volume 32, page ozag007, DOI:10.1093/mam/ozag007.

[49] *2026* **Menglin Zhu***, Michael Xu*, Yubo Qi, Colin Gilgenbach, Jieun Kim, Jiahao Zhang, Bridget R Denzer, Lane W Martin, Andrew M Rappe, and James M LeBeau. Bridging experiment and theory of relaxor ferroelectrics with multislice electron ptychography. *Science*, volume 392, pages 519–523, DOI:10.1126/science.ads6023.

[48] *2026* Yu Yun, Liyan Wu, Drew Behrendt, Hao Pan, **Menglin Zhu**, Michael Xu, Anthony J Ruffino, John Carroll, Irina Baraban, Zishen Tian, Xianfei Xu, Dongfang Chen, Brendan M Hanrahan, Lane W Martin, James M LeBeau, Andrew M Rappe, Ilya Grinberg, and Jonathan E Spanier. Relaxor behavior in rocksalt cation-ordered material induced by (anti)ferroelectric phase competition. *Nat. Commun.*, pages 1–9, DOI:10.1038/s41467-026-74237-z.

[47] *2026* Zishen Tian, **Menglin Zhu**, Jaegyu Kim, Piush Behera, Michael Xu, Thomas J Lee, Ching-Che Lin, Sreekeerthi Pamula, Archana Raja, Hao Pan, Jieun Kim, James M LeBeau, and Lane W Martin. Unleashing the electromechanical response of ferroelastic domain reorganization in mixed-phase tetragonal ferroelectric multilayers. *Adv. Mater.*, page e18417, DOI:10.1002/adma.202518417.

[46] *2026* Colin Gilgenbach, **Menglin Zhu**, and James M LeBeau. Phaser: A unified and extensible framework for fast electron ptychography. *Npj Comput. Mater.*, pages 1–13, DOI:10.1038/s41524-026-01956-8.

[45] *2025* **Menglin Zhu***, Michael Xu*, Yu Yun, Liyan Wu, Or Shafir, Colin Gilgenbach, Lane W Martin, Ilya Grinberg, Jonathan E Spanier, and James M LeBeau. Insights into chemical and structural order at planar defects in Pb2MgWO6 using multislice electron ptychography. *ACS Nano*, DOI:10.1021/acsnano.4c14833.

[44] *2025* **Menglin Zhu***, Michael Xu*, Louis Alaerts, Hao Pan, Colin Gilgenbach, Geoffroy Hautier, Lane W Martin, and James M LeBeau. Direct determination of antiferroelectric-to-ferroelectric phase transition pathways in $PbZrO_3$ with Operando Electron Microscopy. *arXiv [cond-mat.mtrl-sci]*, DOI: arXiv2509.07194.

[43] *2025* **Menglin Zhu***, Joseph Lanier*, Sevim Polat Genlik*, Jose G Flores, Victor da Cruz Pinha Barbosa, Mohit Randeria, Patrick M Woodward, Maryam Ghazisaeidi, Fengyuan Yang, and Jinwoo Hwang. Emergent ferromagnetism at $LaFeO_3/SrTiO_3$ interface arising from a strain-induced spin-state transition. *Adv. Mater. Interfaces*, volume 12, page 2500169, DOI:10.1002/admi.202500169.

[42] *2025* Jith Sarker, Prachi Garg, Abrar Rauf, Ahsiur Rahman Nirjhar, Hsien-Lien Huang, **Menglin Zhu**, A F M Anhar Uddin Bhuiyan, Lingyu Meng, Hongping Zhao, Jinwoo Hwang, Eric Osei-Agyemang, Saquib Ahmed, and Baishakhi Mazumder. Microscopic and spectroscopic investigation of (AlxGa1–X)2O3 films: Unraveling the impact of growth orientation and aluminum content. *Adv. Mater. Interfaces*, volume 12, page 2301016, DOI:10.1002/admi.202301016.

[41] *2025* Hao Pan, Liyan Wu, John Carroll, **Menglin Zhu**, Zishen Tian, Dongfang Chen, Hongrui Zhang, Xianzhe Chen, Xiaoxi Huang, Irina Baraban, Sreekeerthi Pamula, Cedric J G Meyers, R Ramesh, Kathleen Coleman, Brendan Hanrahan, James M LeBeau, Jonathan E Spanier, and Lane W Martin. Highly tunable relaxors developed from antiferroelectrics. *Adv. Mater.*, page e2505376, DOI:10.1002/adma.202505376.

[40] *2025* Jieun Kim, Yubo Qi, Abinash Kumar, Yun-Long Tang, Michael Xu, Hiroyuki Takenaka, **Menglin Zhu**, Zishen Tian, Ramamoorthy Ramesh, James M LeBeau, Andrew M Rappe, and Lane W Martin. Size-driven phase evolution in ultrathin relaxor films. *Nat. Nanotechnol.*, pages 1–9, DOI:10.1038/s41565-025-01863-x.

[39] *2025* I-Hsuan Kao, Junyu Tang, Gabriel Calderon Ortiz, **Menglin Zhu**, Sean Yuan, Rahul Rao, Jiahan Li, James H Edgar, Jiaqiang Yan, David G Mandrus, Kenji Watanabe, Takashi Taniguchi, Jinwoo Hwang, Ran Cheng, Jyoti Katoch, and Simranjeet Singh. Unconventional unidirectional magnetoresistance in heterostructures of a topological semimetal and a ferromagnet. *Nat. Mater.*, pages 1–9, DOI:10.1038/s41563-025-02175-0.

[38] *2025* Deokyoung Kang, Xue-Zeng Lu, Megha Acharya, Sajid Husain, Isaac Harris, Piush Behera, Ching-Che Lin, Ella Banyas, Alex Smith, Francesco Ricci, **Menglin Zhu**, Bridget R Denzer, Tanguy Terlier, Shu Wang, Tae Yeon Kim, Lucas Caretta, Douglas Natelson, James M LeBeau, Jeffrey B Neaton, Ramamoorthy Ramesh, James M Rondinelli, and Lane W Martin. Bulk-Rashba effect with suppressed spin relaxation in a polar phase of Bi1- xIn1+ xO3. *Adv. Mater.*, page e2504684, DOI:10.1002/adma.202504684.

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[36] *2024* **Menglin Zhu**, Joseph Lanier, Jose Flores, Victor da Cruz Pinha Barbosa, Daniel Russell, Becky Haight, Patrick M Woodward, Fengyuan Yang, and Jinwoo Hwang. Structural degeneracy and formation of crystallographic domains in epitaxial $LaFeO_3$ films revealed by machine-learning assisted 4D-STEM. *Sci. Rep.*, volume 14, page 4198, DOI:10.1038/s41598-024-54661-1.

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[19] *2022* I Hsuan Kao, Ryan Muzzio, Hantao Zhang, **Menglin Zhu**, Jacob Gobbo, Sean Yuan, Daniel Weber, Rahul Rao, Jiahan Li, James H Edgar, Joshua E Goldberger, Jiaqiang Yan, David G Mandrus, Jinwoo Hwang, Ran Cheng, Jyoti Katoch, and Simranjeet Singh. Deterministic switching of a perpendicularly polarized magnet using unconventional spin–orbit torques in WTe_2. *Nature Materials 2022 21:9*, volume 21, pages 1029–1034, DOI:10.1038/s41563-022-01275-5.

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[17] *2021* Md Rezaul Karim, Brenton A Noesges, Benthara Hewage Dinushi Jayatunga, **Menglin Zhu**, Jinwoo Hwang, Walter R L Lambrecht, Leonard J Brillson, Kathleen Kash, and Hongping Zhao. Experimental determination of the valence band offsets of $ZnGeN_2$ and $(ZnGe)_{0.94}Ga_{0.12}N_2$$ with GaN. *J. Phys. D Appl. Phys.*, volume 54, page 245102, DOI:10.1088/1361-6463/abee45.

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[7] 2020 A F M Anhar Uddin Bhuiyan, Zixuan Feng, Jared M Johnson, Hsien-Lien Huang, Jith Sarker, **Menglin Zhu**, Md Rezaul Karim, Baishakhi Mazumder, Jinwoo Hwang, and Hongping Zhao. Response to "Comment on 'Phase transformation in MOCVD growth of $(Al_xGa_{1-x})_2O_3$ thin films"'[APL Mater. 8, 089101 (2020)]. *APL Materials*, volume 8, page 89102, DOI:10.1063/5.0014806.

[6] 2020 A F M Anhar Uddin Bhuiyan, Zixuan Feng, Jared M Johnson, Hsien-Lien Huang, Jith Sarker, **Menglin Zhu**, Md Rezaul Karim, Baishakhi Mazumder, Jinwoo Hwang, and Hongping Zhao. Phase transformation in MOCVD growth of $(Al_xGa_{1-x})_2O_3$ thin films. *APL Materials*, volume 8, page 31104, DOI:10.1063/1.5140345.

[5] 2019 Baishakhi Mazumder, Jith Sarker, Yuewei Zhang, Jared M Johnson, **Menglin Zhu**, Siddharth Rajan, and Jinwoo Hwang. Atomic scale investigation of chemical heterogeneity in β-$(Al_xGa_{1-x})_2O_3$ films using atom probe tomography. *Appl. Phys. Lett.*, volume 115, page 132105, DOI:10.1063/1.5113627.

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[3] *2019* Zhiyuan Feng, Belinda Hurley, **Menglin Zhu**, Zi Yang, Jinwoo Hwang, and Rudolph Buchheit. Corrosion Inhibition of AZ31 Mg Alloy by Aqueous Selenite ($SeO_3{}^{2-}$). *J. Electrochem. Soc.*, volume 166, page C520, DOI:10.1149/2.0911914jes.

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Teaching Experience

Fall, 2023 ***3.34 Imaging of Materials***, guest lecturer, Massachusetts Institute of Technology

Fall, 2020 ***MSE3151 Transport Phenomonon and Kinetics***, teaching assistant, Ohio State University

Fall, 2019 ***MSE3332 Undergraduate Lab II***, laboratory assistant and instructor, Ohio State University

Synergistic Activities

2023-present ***Mentor of two graduate students*** on project *Collaborative for Hierarchical Agile and Responsive Materials (CHARM) under cooperative agreement W911NF-19-2-011*

2023 ***Co-organize Microscopy and Microanalysis conference P08 symposium*** entitled *Atomic Scale Microscopy of Interfaces and Heterostructures with Correlated Phenomena*

2023 ***Mentor of two REU Students*** on porject *NFO Thin Films Grown via an Off-Axis Sputtering Method; STEM Characterization of $NdFeO_3$/$SrTiO_3$ Thin Films*

2022 ***Mentor of one REU Student*** on porject *Effects of Octahedral Tilting and Lattice Strain on $LaFeO_3$*

References

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